

115-1 FUNDAMENTALS OF BIOINFORMATICS

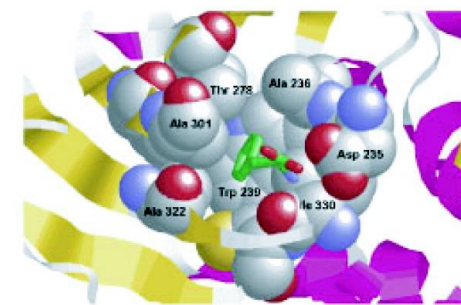
Chapter 5: How Do We Compare Biological Sequences?

<https://youtube.com/playlist?list=PLQ-85IQIPqFNmbPEsMoxb5dM5qtRaVShn&si=cGoWIDJUfrKkTFil>

What is a domain?

- Adenylation domain
- Homeodomain

(a)



(b)

GenA-M1-Phe QPARISF...
Cda1-M2-Thr LPREYAP...
FanD-M2-Thr LPREYCT...
BacC-M3-Asn LYSSIAF...
CepA-M3-Asn MHSIAF...



Cda1-M2-Thr D F W N V G M V
FanD-M2-Thr D F W N I G M V
BacC-M3-Asn D L T K I G R V
CepA-M3-Asn D L T K L G R V

Chemistry & Biology

Blue Positions in A-domains Define Non-Ribosomal Code

YAFDLGYTCMFPV...
-AFDVSA...
IAPDASSWEIYAP...

LTKVGHIG Asp
VGEIGSID Orn
AWMFAAVL Val

[Predictive, structure-based model of amino acid recognition by nonribosomal peptide synthetase adenylation domains - ScienceDirect](#)

What is a domain?

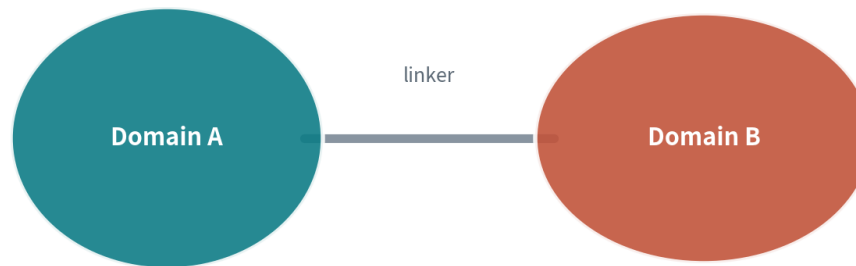
Protein domain

A polypeptide chain folds into several independently-folding, functional units

(1) Domain architecture: draw each domain as a coloured segment



(2) In 3-D: each domain folds into a compact globular unit, joined by a linker

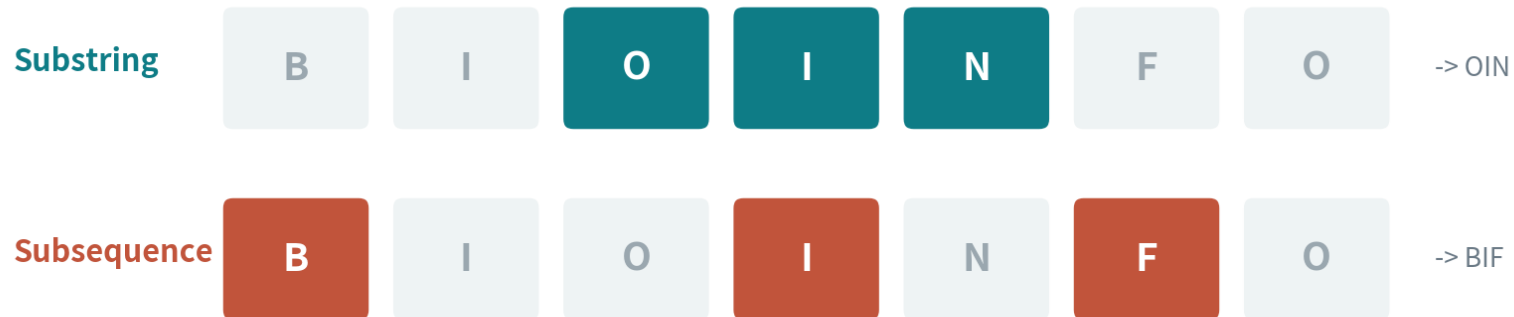


What is the difference between subsequences and substrings?

- String
 - Subsequence
 - Substring
-
- Common subsequence
 - Longest common subsequence problem

What is the difference between subsequences and substrings?

Subsequence vs. Substring



A substring must be contiguous; a subsequence only needs to keep the relative order.

In order

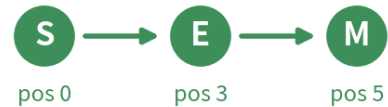
In order

'In order': a subsequence keeps the original left-to-right order



Take the 3 highlighted letters and read them out in some order:

read "SEM"



positions $0 < 3 < 5$ (increasing) = a subsequence

read "MES"



positions $5 > 3 > 0$ (decreasing) = NOT a subsequence

Where is the error? Reading M-E-S needs positions 5,3,0 - the indices go backwards. You may skip letters, but never reorder them.

Original figure · CC0

indel

- Insertion
- Deletion

indel

indel = insertion / deletion

Deletion

A	C	G	T	A	C
---	---	---	---	---	---

A	C	-	T	A	C
---	---	---	---	---	---

present in top, missing in bottom -> a deletion (shown as gap '-')

Insertion

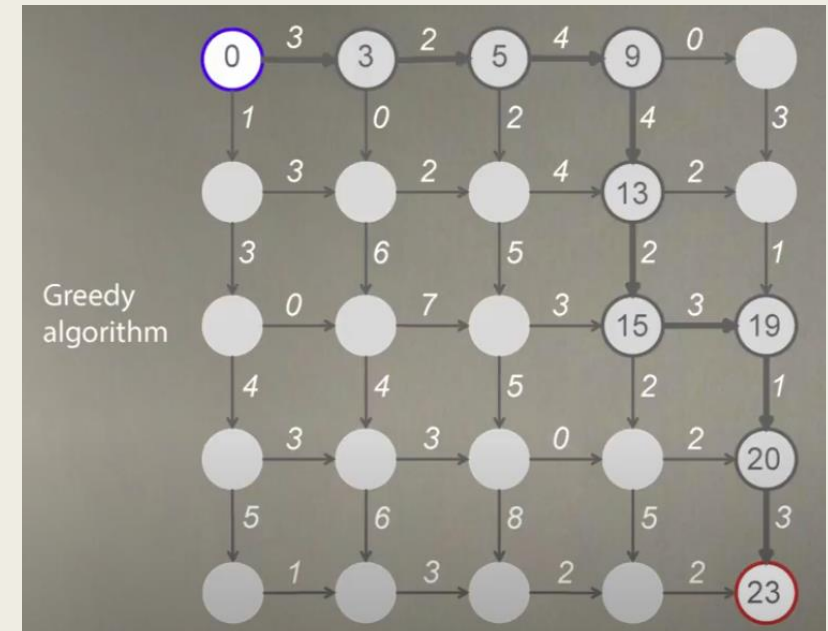
A	C	-	T	A	C
---	---	---	---	---	---

A	C	G	T	A	C
---	---	---	---	---	---

bottom has an extra character -> an insertion; together they are called indels

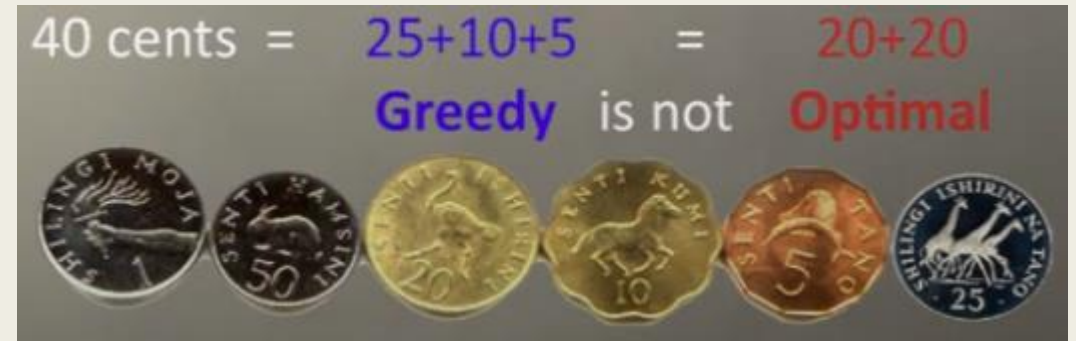
Manhattan Tourist Problem

- Longest path from the source to the sink in the grid
- Strategy:
 - *Greedy algorithm*
- Optimal?



The Change Problem

- Find the minimum number of coins needed to make change



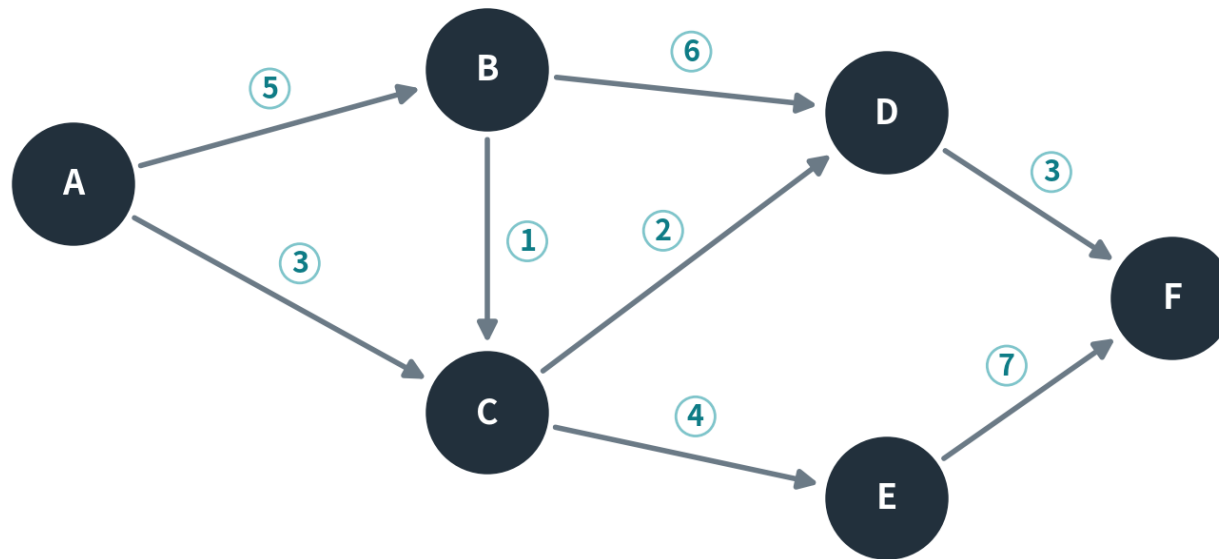
- Recursive Change
- Dynamic programming vs. Recursion

<https://youtu.be/Aqu9Zd6qqQk?si=zDJJZHVjuio5xYDk>

Edge-weighted directed graph

Edge-weighted directed graph

Edge-weighted directed graph

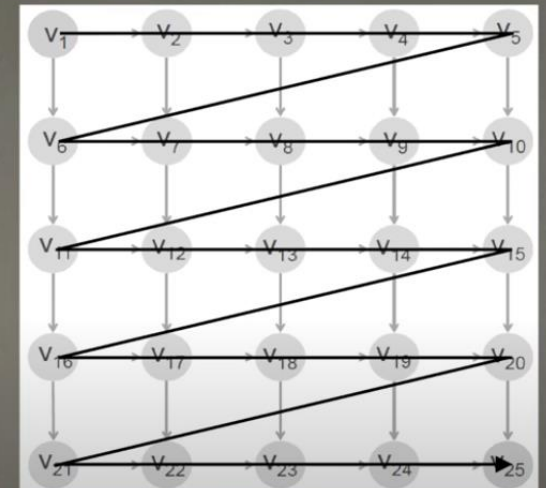
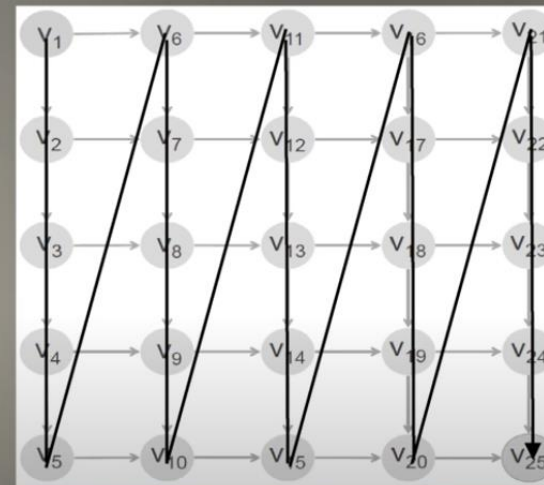


Nodes = vertices; arrows = directed edges; numbers = edge weights (a cost or score)

DAG

- Topological order
- How many topological orderings of Manhattan grid?

Two Topological Orderings of the Manhattan Grid



Big-O

- Running time is proportional to
 - *The number of edges?*

LongestPath

LongestPath(*Graph*, *source*, *sink*)

for each node *a* in *Graph*

$s_a \leftarrow -\text{infinity}$

$s_{\text{source}} \leftarrow 0$

topologically order *Graph*

for each node *a* (from *source* to *sink* in the topological order)

$s_a \leftarrow \max_{\text{all predecessors } b \text{ of node } a} \{s_b + \text{weight of edge from } b \text{ to } a\}$

return s_{sink}

runtime: $O(\text{\#edges})$ in the graph

Scoring matrix

- Please find some scoring (substitution) matrices for
 - *amino acids*
 - *nucleotides*

Scoring matrix

Scoring / substitution matrix

Amino acids: BLOSUM62 (subset)

	A	R	N	D	C	Q
A	4	-1	-2	-2	0	-1
R	-1	5	0	-2	-3	1
N	-2	0	6	1	-3	0
D	-2	-2	1	6	-3	0
C	0	-3	-3	-3	9	-3
Q	-1	1	0	0	-3	5

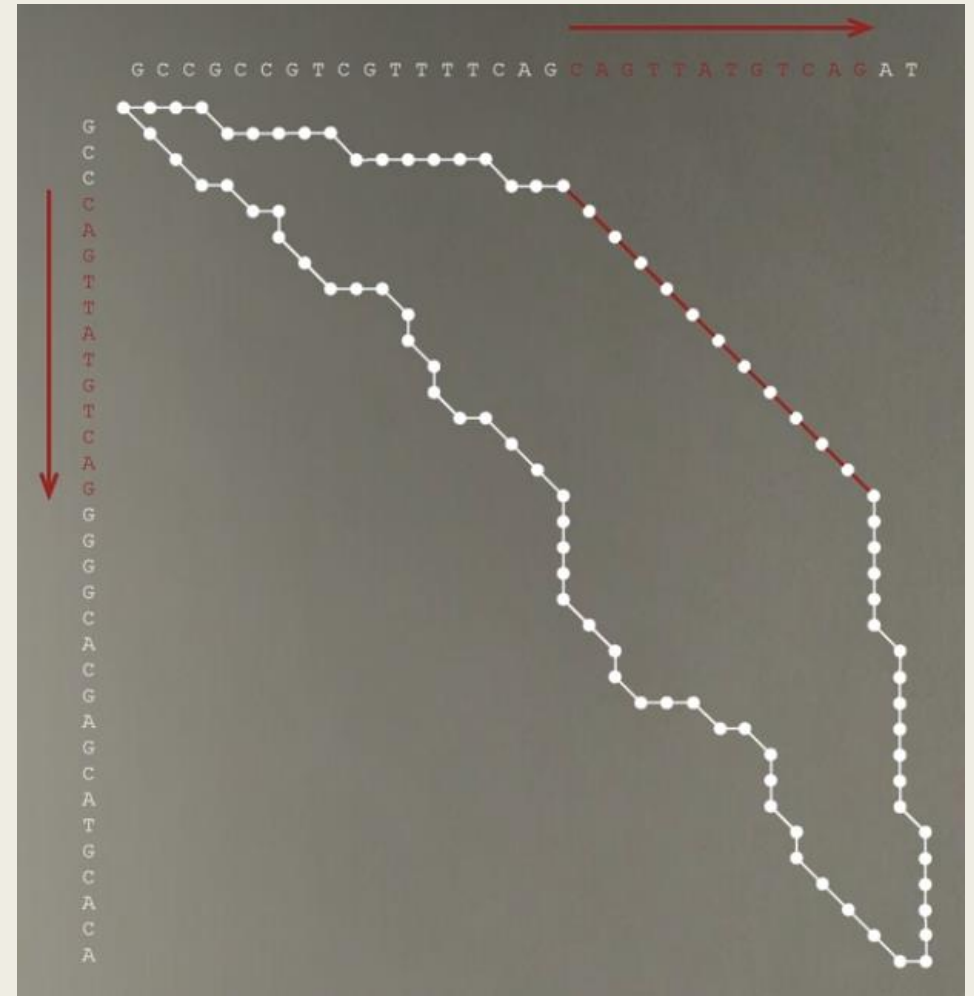
Nucleotides: match +1 / mismatch -

	A	C	G	T
A	+1	-1	-1	-1
C	-1	+1	-1	-1
G	-1	-1	+1	-1
T	-1	-1	-1	+1

Positive = favourable/conservative swap; negative = rare; diagonal (same residue) is highest

Global alignment vs. local alignment

- entire



Smith-waterman algorithm

Smith-Waterman algorithm

Smith-Waterman: local alignment

		G	A	T	T	A	C	A
		0	0	0	0	0	0	0
G		0	2	1	0	0	0	0
T		0	1	1	3	2	1	0
T		0	0	0	3	5	4	3
A		0	0	2	2	4	7	6
C		0	0	1	1	3	6	9

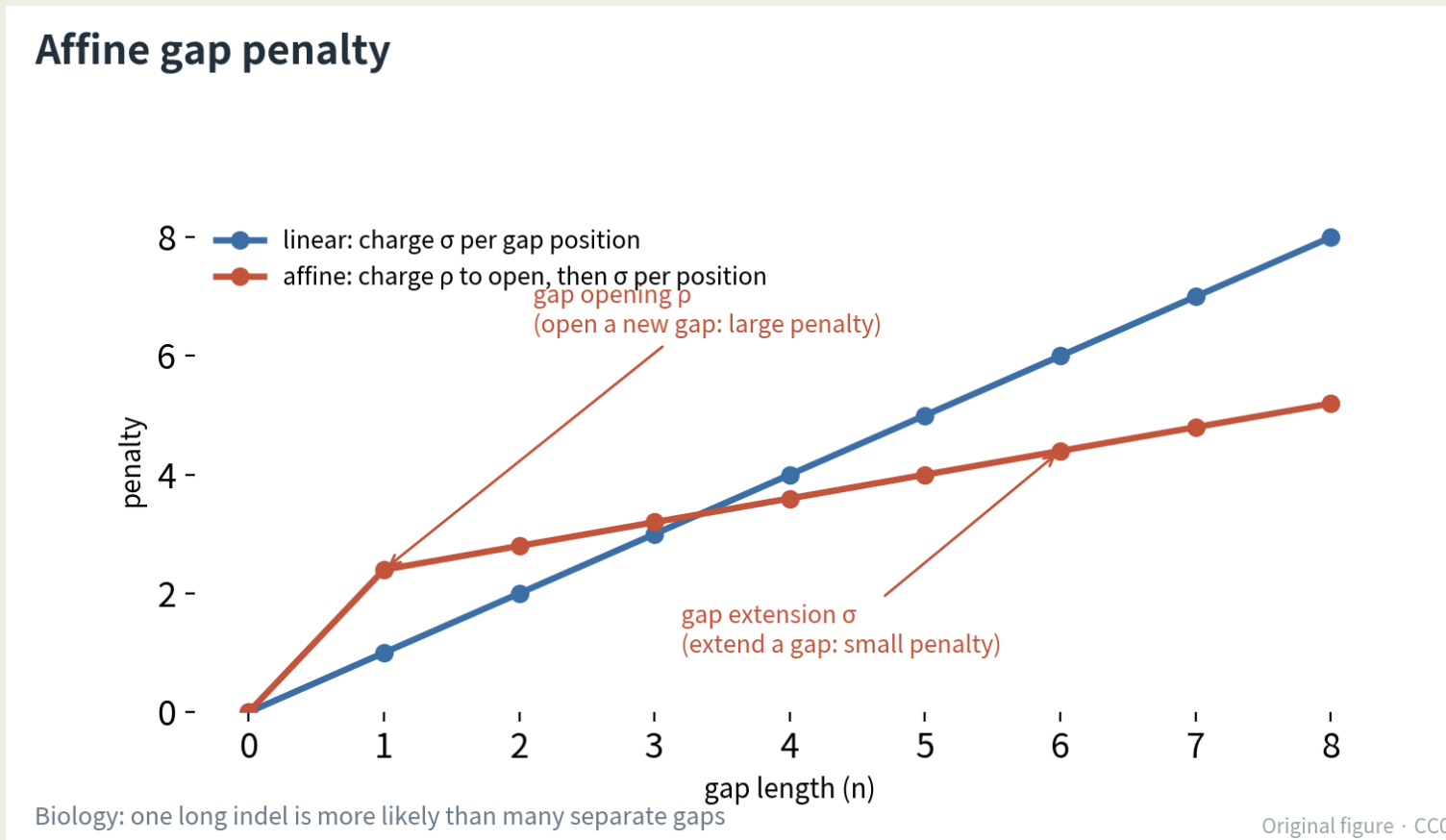
Negative scores reset to 0, keeping only local high-scoring regions; trace back from the top cell (gold) for the best local alignment.

Compare: global alignment always spans both full sequences (top-left to bottom-right)

Original figure · CC0

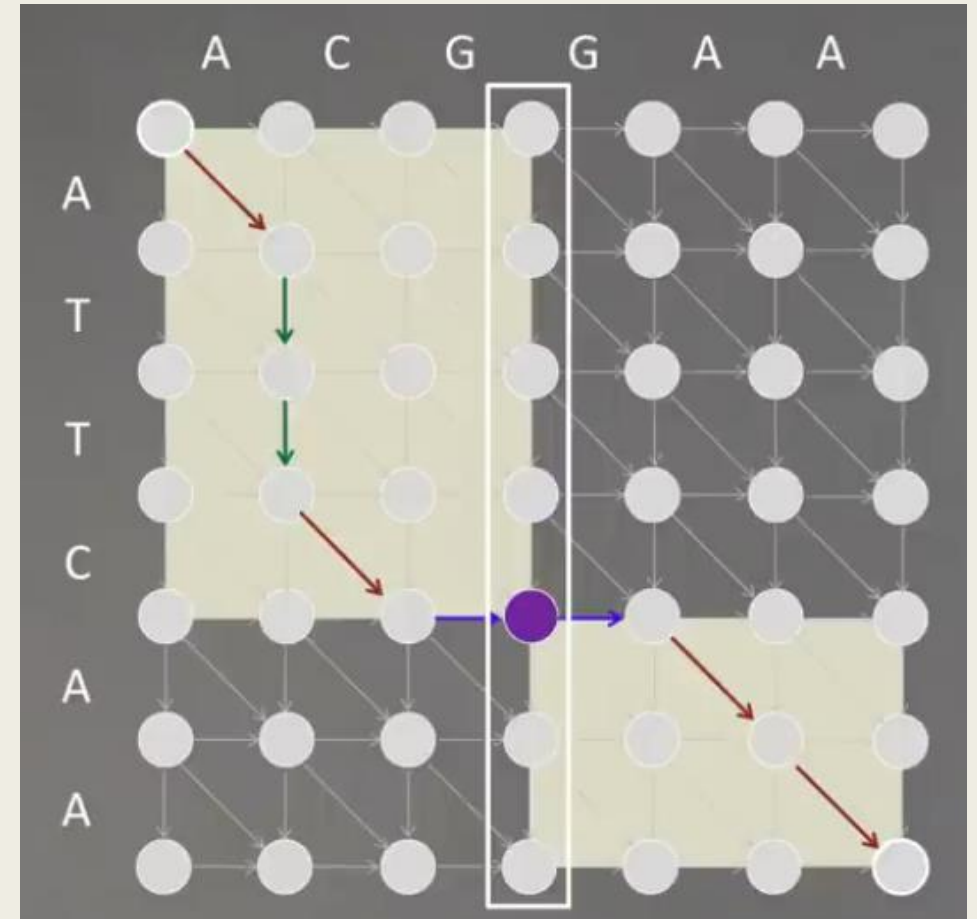
Affine gap penalty

Affine gap penalty



Space-efficient

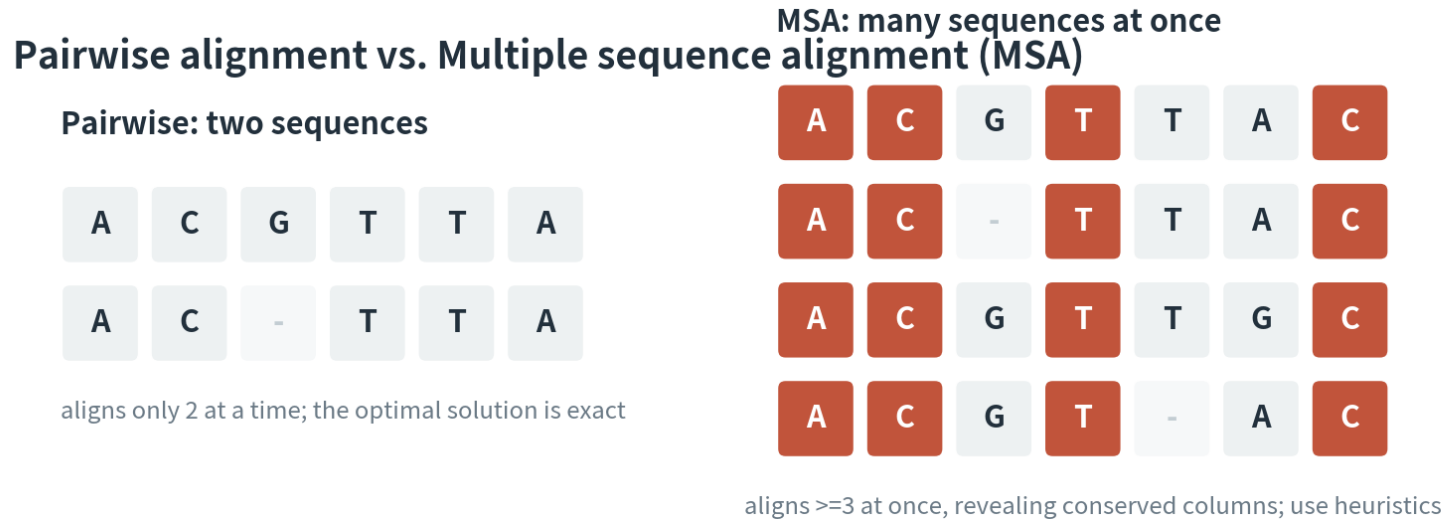
- In linear space



Pairwise alignment vs. multiple sequence alignment

- Heuristic algorithms

Pairwise alignment vs. multiple sequence alignment



Profile representation

- PSSM
- Profile alignment

Profile representation

Profile / PSSM and sequence logo

(1) A set of aligned sequences

A	T	G	C	A
A	T	G	C	A
A	T	G	T	A
A	C	G	C	A
A	T	G	C	G

(2) As a sequence logo: letter height $\propto$ column conservation



more conserved (more consistent) $\rightarrow$ taller letters; a PSSM scores each

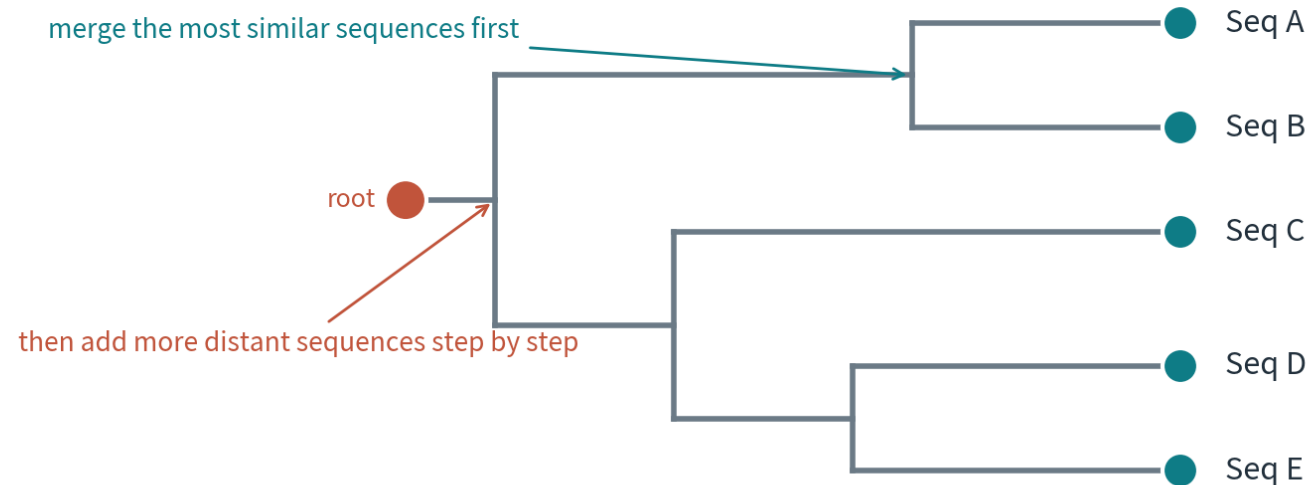
Original figure - CC0

Progressive alignment

- Phylogenetical tree

Progressive alignment

Guide tree for progressive alignment



Build a tree by similarity, then add sequences progressively from closest to farthest to build the MSA

Original figure · CC0